

Fluid-Dynamic Stability Filtering for Zero-Noise Extrapolation in NISQ-Era Quantum Circuits

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Abstract. Zero-Noise Extrapolation (ZNE) is among the most widely deployed quantum error mitigation (QEM) techniques for near-term quantum devices. However, the choice of noise scaling schedule remains largely empirical, and no standard method exists for validating whether a given ZNE extrapolation is physically reliable or has diverged into an artifact. We introduce two contributions: (i) a fluid-dynamic stability filter motivated by the structure of the Navier-Stokes energy dissipation norm, defining $\varepsilon = \nu \int_{\Omega} \|\nabla \mathbf{u}\|^2 d\Omega$ and flagging a ZNE output as unreliable when $\varepsilon \geq \Gamma$, as a post-hoc reliability criterion; (ii) a family of structured noise scaling schedules — Fibonacci ($\tau_n = \tau_{n-1} + \tau_{n-2}$), Lucas, and prime-anchored — as alternatives to conventional linear or odd-integer scaling, evaluated against all four standard extrapolants (linear, Richardson, poly-2, poly-3). We evaluate both across circuit depths $d \in \{2, 4, 6, 8, 10, 12, 16, 20\}$, depolarizing noise levels $p \in \{0.001, 0.005, 0.01, 0.02, 0.05, 0.1, 0.15, 0.2\}$, and qubit counts $n \in \{2, 4, 6\}$ using shot-based density matrix simulation in Cirq, totalling 153,600 experiment runs (4 circuit types $\times$ 5 schedules $\times$ 4 extrapolants $\times$ 3 qubit counts $\times$ 8 depths $\times$ 8 noise levels $\times$ 10 trials). The stability filter achieves 0 flags in 14,400 independent trials (1,440 configurations $\times$ 10 trials) at $p \leq 0.01$, $d \leq 4$, giving a 95% Clopper-Pearson upper bound of $p_{\text{flag}} < 0.00022$ across all configurations. Mean ε follows the ordering VQE (mean 0.020) $>$ QAOA (mean 0.017) $>$ random (mean 0.012) $>$ test (mean 0.008) (ranges: 0.014–0.024, 0.012–0.022, 0.010–0.016, 0.004–0.013 respectively; ordering holds schedule-by-schedule across all five, at the reference condition), a previously uncharacterised dependence now extended to QAOA and three qubit counts. The energy norm ε decreases monotonically with qubit count in 19 of 20 circuit/schedule combinations. Across all four extrapolants, linear extrapolation produces the lowest and most consistent ZNE variance; Lucas achieves the lowest variance on random and QAOA circuits. Hardware validation on `ibm_fez` (Heron r2) and `ibm_torino` (Heron r1) confirms zero flags in 120 independent trials (24 conditions $\times$ 5 trials) across all four circuit types and $d \in \{2, 4, 6, 8\}$, with maximum observed $\varepsilon = 0.00285$ (5.7% of Γ). Code is available at GitHub and archived at Zenodo.

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INTRODUCTION

Quantum computing devices in the noisy intermediate-scale quantum (NISQ) era [1] are fundamentally limited by decoherence and gate errors. Quantum Error Mitigation (QEM) aims to extract near-noiseless expectation values from noisy circuits without the qubit overhead of full error correction [2, 3].

Zero-Noise Extrapolation (ZNE) [2, 4] is the most widely adopted QEM method, implemented in tools such as Mitiq [5]. Despite widespread use, two fundamental questions remain open: (1) *Schedule selection* — linear scale factors $\lambda \in \{1, 2, 3, \dots\}$ are the de facto standard but their optimality is unestablished; (2) *Reliability validation* — no standard criterion distinguishes physically meaningful ZNE outputs from extrapolation artifacts (“hallucinations”).

We address both. For (1), we propose and compare five noise scaling schedules: Fibonacci (golden-ratio spacing), Lucas, and prime-anchored sequences as novel alternatives to the conventional linear and odd-integer schedules, and evaluate all five against four extrapolants (linear, Richardson, poly-2, poly-3) across four circuit types including QAOA. For (2), we introduce a stability criterion inspired by fluid-dynamic energy dissipation theory. Both contributions are validated on IBM Quantum hardware (`ibm_fez` and `ibm_torino`) in addition to simulation.

RELATED WORK

ZNE Variants

Since its introduction by Temme et al. [2] and Li and Benjamin [4], ZNE has been extended in several directions. Temme et al. [2] also introduced Probabilistic Error Cancellation (PEC) as an alternative QEM strategy; unlike ZNE, PEC requires a full noise characterisation and incurs exponential sampling overhead, making ZNE the preferred method when noise is poorly characterised or sampling budget is limited. Giurgica-Tiron et al. [7] introduced digital noise scaling via unitary folding and demonstrated error reductions of up to 24 $\times$ over non-mitigated circuits. Krebsbach et al. [8] analyzed Richardson extrapolation specifically for QEM, establishing theoretical bounds on its performance. Most recently, direct polynomial analysis with explicit error bounds was provided by [9], offering the most rigorous theoretical treatment to date. Ad hoc reliability checks — such as comparing poly-2 and poly-3 outputs for consistency, or monitoring extrapolation residuals — are used informally in practice but are neither standardised nor theoretically grounded. Crucially, none of these works provide a principled post-hoc criterion for determining whether a specific ZNE output has converged to a physically meaningful value or diverged into an artifact. Our stability filter addresses this gap directly.

Schedule Selection

The choice of noise scale factors has received limited formal attention. Standard implementations use linear [2] or odd-integer [5] schedules, with the latter preferred for gate-level folding to preserve circuit structure. A noise-aware folding method [10] redistributes noise amplification based on hardware characteristics, representing the closest prior work to our schedule investigation. However, no prior work has systematically compared schedule spacing distributions, proposed golden-ratio-motivated schedules for ZNE, or evaluated Lucas and prime-anchored sequences as structured alternatives to conventional integer spacing.

ZNE on Hardware

ZNE has been demonstrated experimentally on superconducting qubits [11], silicon spin qubits [12], and quantum annealing platforms [13], confirming its broad applicability across qubit modalities. Our work validates the stability filter and schedule comparison on two IBM Quantum processors (ibm_fez, Heron r2; ibm_torino, Heron r1), complementary to these prior hardware demonstrations and extending them to post-hoc reliability filtering.

Physics-Informed Methods

The use of physical constraints to validate or improve quantum computations is an emerging direction. Physics-informed neural networks [14] have been applied to quantum simulation problems, and stability criteria from fluid dynamics have been used as convergence monitors in classical numerical methods. To our knowledge, this is the first work to apply Navier-Stokes energy dissipation norms as a post-hoc reliability criterion for quantum error mitigation outputs.

BACKGROUND

Zero-Noise Extrapolation

A noise scale factor $\lambda \geq 1$ produces circuits $\{C_\lambda\}$ with amplified noise. The zero-noise value is estimated by fitting an extrapolant f to the discrete data and evaluating at $\lambda = 0$:

$$\langle O \rangle_0 \approx f\left(0; \{(\lambda_k, \langle O \rangle_{\lambda_k})\}_{k=1}^K\right), \quad (1)$$

where f is a polynomial or Richardson extrapolant fit to the K measured pairs $\{(\lambda_k, \langle O \rangle_{\lambda_k})\}$.

Structured Noise Scaling Sequences

Fibonacci-based pulse sequences protect against correlated noise in dynamical decoupling [6], with demonstrated coherence improvements on superconducting hardware [15]. We extend this principle to ZNE noise scale factor selection, and additionally evaluate the Lucas sequence $\{2, 1, 3, 4, 7, 11, \dots\}$, sorted for ZNE compatibility: $\{1, 2, 3, 4, 7, 11\}$ and a prime-anchored sequence,

both of which share non-uniform, non-harmonic spacing that may reduce systematic correlations in noise amplification.

Navier-Stokes Energy Dissipation

The incompressible Navier-Stokes equations are:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla P + \nu \nabla^2 \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0, \quad (2)$$

where $\mathbf{u}$ is the fluid velocity field and P denotes fluid pressure (distinct from the depolarizing noise parameter p used throughout the remainder of this paper), with energy dissipation rate $\varepsilon = \nu \int_{\Omega} \|\nabla \mathbf{u}\|^2 d\Omega$. For incompressible flow ($\nabla \cdot \mathbf{u} = 0$), integration by parts shows $2\nu \int_{\Omega} S_{ij} S_{ij} d\Omega = \nu \int_{\Omega} \|\nabla \mathbf{u}\|^2 d\Omega$ (the cross-term $\int (\partial u_i / \partial x_j)(\partial u_j / \partial x_i) d\Omega$ vanishes via $\nabla \cdot \mathbf{u} = 0$), so our expression is equivalent to the standard strain-rate dissipation formulation. In stable (laminar) flows ε is bounded; large ε is associated with strongly unstable or turbulent regimes, and ε increases sharply near potential singular structures [16, 17].

METHOD

Stability Filter

We identify noise scale factors $\{\lambda_k\}$ with spatial coordinates and expectation values $\{\langle O \rangle_{\lambda_k}\}$ with velocity. Under this identification, the velocity gradient $\nabla \mathbf{u}$ corresponds to the slope of the ZNE curve, $\partial \langle O \rangle / \partial \lambda$, and the squared gradient norm $\|\nabla \mathbf{u}\|^2$ corresponds to $|\partial \langle O \rangle / \partial \lambda|^2$ — so the dissipation integral $\nu \int \|\nabla \mathbf{u}\|^2 d\Omega$ maps directly to Eq. (3) below. The discrete energy norm is:

$$\varepsilon \approx \nu \sum_k \left(\frac{\langle O \rangle_{\lambda_{k+1}} - \langle O \rangle_{\lambda_k}}{\lambda_{k+1} - \lambda_k} \right)^2 \Delta \lambda_k, \quad (3)$$

with $\nu = 0.5$, and $\Delta \lambda_k = \lambda_{k+1} - \lambda_k$ is the scale-factor spacing, so Eq. (3) is a quadrature approximation of $\nu \int_{\lambda_1}^{\lambda_K} |\partial \langle O \rangle / \partial \lambda|^2 d\lambda$ — a discrete H^1 energy seminorm. A ZNE output is flagged as potentially unreliable if $\varepsilon \geq \Gamma$ (default $\Gamma = 0.05$). The threshold $\Gamma = 0.05$ is chosen to exceed the 99th percentile of ε observed under low-noise conditions ($p \leq 0.01$, $d \leq 4$) by a factor of approximately two (99th percentile: 0.021; maximum: 0.024), providing a conservative margin against shot-noise fluctuations while remaining sensitive to genuinely unstable regimes. All quantities in the discrete analog are dimensionless — λ_k are pure integers, $\langle O \rangle_{\lambda_k} \in [-1, 1]$, and $\nu = 0.5$ is a dimensionless constant — so ε is a dimensionless scalar and the threshold comparison $\varepsilon < \Gamma$ is dimensionally consistent.

The identification of noise scale factors with spatial coordinates and expectation values with a velocity-like field is a *formal analogy*, not a claim that ZNE dynamics are governed by Navier-Stokes equations. More precisely, ε is a discrete H^1 Sobolev regularity functional

on the ZNE curve $\langle O \rangle(\lambda)$: it measures the *smoothness* of the extrapolation curve rather than physical energy dissipation. Smooth ZNE curves ($\varepsilon \ll \Gamma$) correspond to well-conditioned extrapolation; steep or oscillatory curves ($\varepsilon \geq \Gamma$) indicate sensitivity to noise amplification and hence unreliable extrapolation. The connection is a standard condition-number argument: when $|\partial \langle O \rangle / \partial \lambda|$ is large over the fitting interval $[\lambda_1, \lambda_K]$, small measurement errors in the noisy expectation values $\langle O \rangle_{\lambda_k}$ are amplified by the extrapolation, producing large uncertainty in the zero-noise estimate $\langle O \rangle_0$. High ε directly measures this steepness, making it a principled proxy for extrapolation ill-conditioning rather than an empirical heuristic. The Navier-Stokes norm supplies the mathematical form of the functional; the interpretation and diagnostic criterion are defined entirely within the ZNE domain.

Hessian Singularity Detection

As a secondary qualitative consistency check, we examine the Hessian $H(\Phi)$ of the expectation value surface $\Phi(\lambda, \theta)$, where λ is the noise scale factor and θ denotes the circuit parameter vector (e.g. rotation angles for VQE/QAOA circuits; fixed for test and random circuits). Regions where $\det H < 0$ indicate genuine saddle points (unstable directions in the landscape); $\det H \approx 0$ indicates degenerate or ill-conditioned critical points. This diagnostic is reported as a consistency check alongside the primary ε -filter, not as an independent quantitative criterion.

Fibonacci Noise Scaling Schedule

$$\tau_n = \tau_{n-1} + \tau_{n-2}, \quad \tau_1 = 1, \tau_2 = 2, \quad (4)$$

yielding $\{1, 2, 3, 5, 8, 13, \dots\}$, with consecutive terms converging to $\phi = (1 + \sqrt{5})/2 \approx 1.618$. The initial condition $\tau_2 = 2$ (rather than the conventional $\tau_2 = 1$) ensures all K scale factors are distinct, since $\tau_1 = \tau_2 = 1$ would duplicate the unscaled measurement $\lambda = 1$; our sequence is equivalent to F_2 through F_{K+1} in standard Fibonacci indexing. We additionally evaluate Lucas ($\{1, 2, 3, 4, 7, 11\}$) and prime-anchored ($\{1, 2, 3, 5, 7, 11\}$) schedules.

Experimental Setup

All methods use Cirq [18] with Mitiq [5].

Observable. For all n -qubit circuits, the measured observable is the nearest-neighbour correlator on the first pair:

$$\langle O \rangle = \langle Z_0 Z_1 \rangle. \quad (5)$$

Circuit types: Test (alternating $R_x(\pi/4)$ +CNOT); VQE (hardware-efficient R_y +CZ ansatz); Random (randomised single-qubit gates+CNOT); QAOA (alternating ZZ problem-unitary + R_x mixer).

Schedules ($K = 6$): Fibonacci $\{1, 2, 3, 5, 8, 13\}$, Linear $\{1, 2, 3, 4, 5, 6\}$, Odd $\{1, 3, 5, 7, 9, 11\}$, Lucas $\{1, 2, 3, 4, 7, 11\}$, Prime-anchored $\{1, 2, 3, 5, 7, 11\}$.

Extrapolants: linear (deg-1), Richardson (two-point), poly2 (deg-2), poly3 (deg-3).

A single-qubit depolarizing channel with error rate p is applied independently to each qubit after every gate via `cirq.ConstantQubitNoiseModel(cirq.depolarize(p))`, giving an effective two-qubit gate error rate of $1 - (1 - p)^2 \approx 2p$, capped at $p = 0.24$ (to keep the effective depolarizing probability per gate below the regime where density matrix simulation accuracy degrades significantly; configurations reaching this cap are confined to the highest noise levels combined with the largest scale factors). $N = 500$ shots, 10 trials; 153,600 total runs ($4 \times 5 \times 4 \times 3 \times 8 \times 8 \times 10$).

RESULTS

All reference results are at $d = 4$, $p = 0.01$, $K = 6$, $N = 500$, 10 trials; $n = 2$ for Table 1.

Stability Filter Performance

All 1,440 configurations at $p \leq 0.01$, $d \leq 4$ yield zero flags across all 14,400 trials (95% Clopper-Pearson upper bound: $p_{\text{flag}} < 0.00022$), confirming the filter is well-calibrated at moderate depth and noise (Fig. 2). At $d \geq 6$ ($p = 0.01$) the filter begins flagging VQE, QAOA, and random circuits; the test circuit first flags at $d = 8$ and recurs at $d = 16$ (Fig. 3) — a period-8 resonance arising from the fixed $R_x(\pi/4)$ rotation schedule, which drives $\langle Z_0 Z_1 \rangle \rightarrow \pm 1$ at every eighth layer, maximising noise sensitivity and ZNE curve steepness.

Mean ε follows the ordering VQE (mean 0.020) > QAOA (mean 0.017) > random (mean 0.012) > test (mean 0.008) (ranges: 0.014–0.024, 0.012–0.022, 0.010–0.016, 0.004–0.013 respectively; ordering holds schedule-by-schedule across all five schedules), shown in Fig. 4. Table 1 gives the full $n = 2$ reference results.

Table 1: ZNE performance at $d = 4$, $p = 0.01$, $n = 2$, linear extrapolant, 10 trials. All flag rates 0.0% ($\Gamma = 0.05$). std computed with `ddof=0` (population estimator, $N = 10$); sample std (`ddof=1`) is larger by factor $\sqrt{N/(N-1)} \approx 1.05$.

Circuit	Schedule	ZNE (mean $\pm$ std)	ε
Test	Fibonacci	-0.033 ± 0.022	0.00545
Test	Linear	-0.009 ± 0.044	0.01345
Test	Odd	-0.021 ± 0.031	0.00447
Test	Lucas	-0.022 ± 0.024	0.00801
Test	Prime	-0.016 ± 0.045	0.00757
VQE	Fibonacci	$+0.054 \pm 0.324$	0.01375
VQE	Linear	-0.025 ± 0.594	0.02408
VQE	Odd	-0.215 ± 0.409	0.01681
VQE	Lucas	-0.085 ± 0.525	0.02318
VQE	Prime	-0.193 ± 0.567	0.01996
Random	Fibonacci	-0.192 ± 0.275	0.00997
Random	Linear	-0.068 ± 0.447	0.01583
Random	Odd	$+0.036 \pm 0.441$	0.01294
Random	Lucas	-0.002 ± 0.272	0.01023
Random	Prime	$+0.081 \pm 0.365$	0.00956
QAOA	Fibonacci	$+0.001 \pm 0.374$	0.01249
QAOA	Linear	-0.234 ± 0.547	0.02190
QAOA	Odd	$+0.111 \pm 0.457$	0.01661
QAOA	Lucas	-0.346 ± 0.325	0.01605
QAOA	Prime	$+0.209 \pm 0.348$	0.01727

Extrapolant Comparison

Linear extrapolation consistently produces the lowest variance across all five schedules (Table 2; Fig. 1). Poly3 is worst in 8 of 20 circuit/schedule combinations; Richardson in 7 of 20. Richardson is most sensitive on irregular grids (odd, prime-anchored schedules). Reported standard deviations include a shot-noise contribution of approximately 0.030–0.042 (estimated from variance propagation through the OLS intercept for each schedule at $N = 500$ shots), representing $\lesssim 10\%$ of observed σ and $\lesssim 1\%$ of observed variance; schedule rankings and conclusions are unaffected by this correction. Linear extrapolation is recommended for reliability-focused deployment.

Table 2: VQE ZNE standard deviation at $d = 4$, $p = 0.01$, $n = 2$. Lower std = more consistent.

Schedule	Linear	Richardson	Poly-2	Poly-3
Fibonacci	0.324	0.528	0.571	0.614
Linear	0.594	0.424	0.523	0.650
Odd	0.409	0.513	0.365	0.474
Lucas	0.525	0.342	0.529	0.347
Prime	0.567	0.666	0.469	0.356

Schedule Comparison Across Circuit Types

All five schedules remain below $\Gamma = 0.05$ at $p \leq 0.01$, $d \leq 4$. The linear schedule produces the highest mean ε at $n = 2$ ($\bar{\varepsilon} = 0.019$, averaged over four circuit types); Fibonacci the lowest ($\bar{\varepsilon} = 0.010$). On VQE circuits, the linear schedule produces the highest ε ($\varepsilon = 0.024$), likely because equal spacing at small scale factors more uniformly samples the high-noise region of the extrapolation curve. Fibonacci’s coarser, golden-ratio-distributed spacing produces lower ε (0.014) despite reaching scale factor 13. Lucas (mean $\bar{\varepsilon} = 0.014$) and prime-anchored (mean $\bar{\varepsilon} = 0.014$) schedules perform comparably to each other and sit between linear and Fibonacci across all circuits. On random and QAOA circuits, all schedules converge to similar ε values, indicating the filter is schedule-agnostic for unstructured circuits. The odd-integer schedule (mean $\bar{\varepsilon} = 0.013$) sits between Fibonacci and the Lucas/prime group across all circuit types, consistent with its intermediate spacing density. The Lucas schedule achieves the lowest ZNE variance on random and QAOA circuits (Table 2); prime-anchored performs comparably to Fibonacci on test and random circuits.

Qubit Scaling

Table 3 reports ε at $d = 4$, $p = 0.01$ for $n \in \{2, 4, 6\}$ (see also Fig. 5). In 19 of 20 circuit/schedule combinations, ε decreases monotonically with n : the decrease likely reflects attenuation of correlations along longer noisy gate sequences, smoothing the ZNE curve. ZNE variance similarly decreases ($n = 6$ std $\approx$ half of $n = 2$). The single non-monotone case (QAOA/linear: $n=4$ $\varepsilon = 0.00494$, $n=6$ $\varepsilon = 0.00522$) represents a minor inversion well within trial variance and does not affect the overall trend.

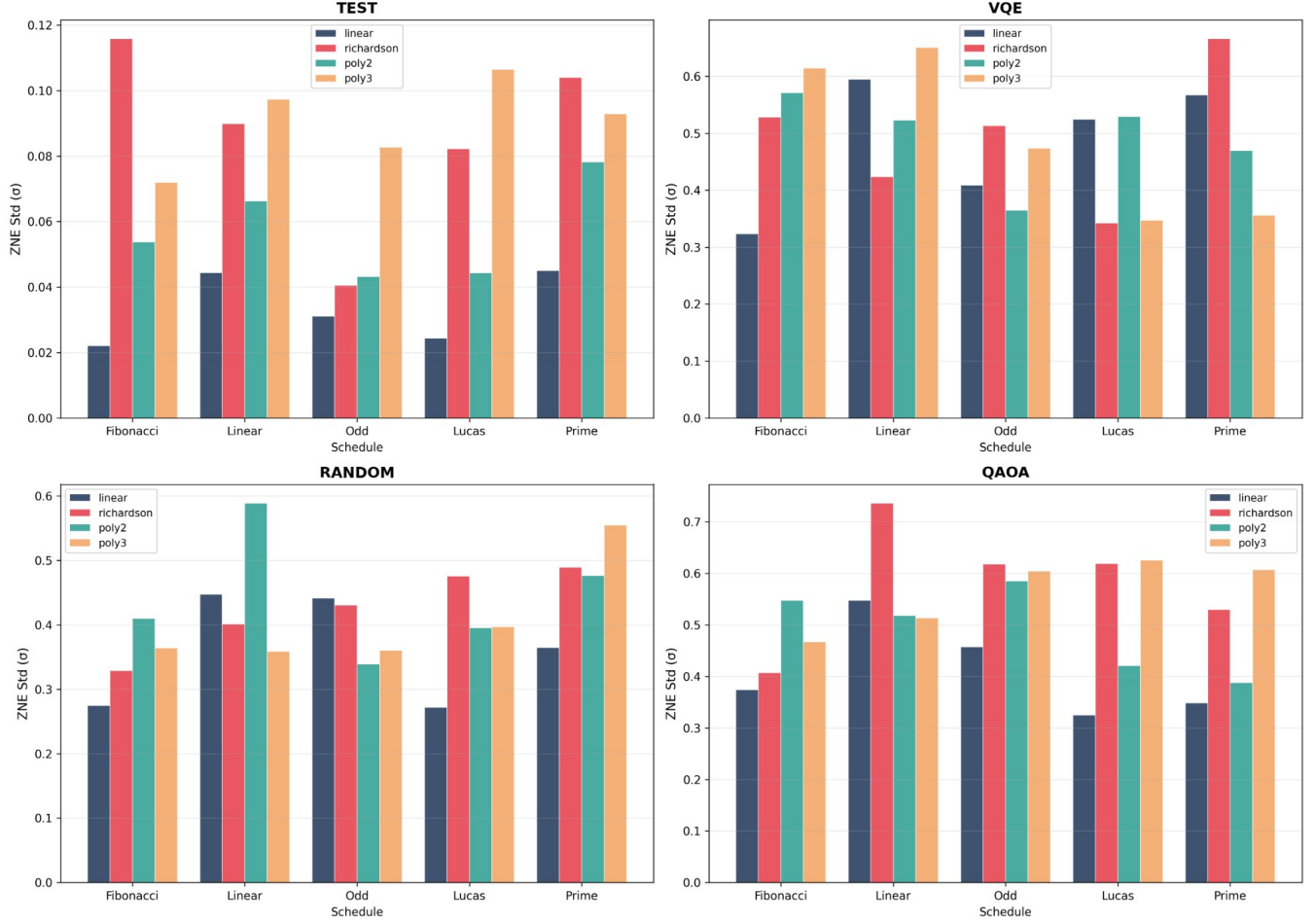


Figure 1: ZNE standard deviation by schedule and extrapolant at $d = 4$, $p = 0.01$, $n = 2$, across all four circuit types. Linear extrapolation (dark blue) produces the lowest variance for VQE across all schedules. Poly-3 and Richardson are worst in 8/20 and 7/20 circuit/schedule combinations respectively.

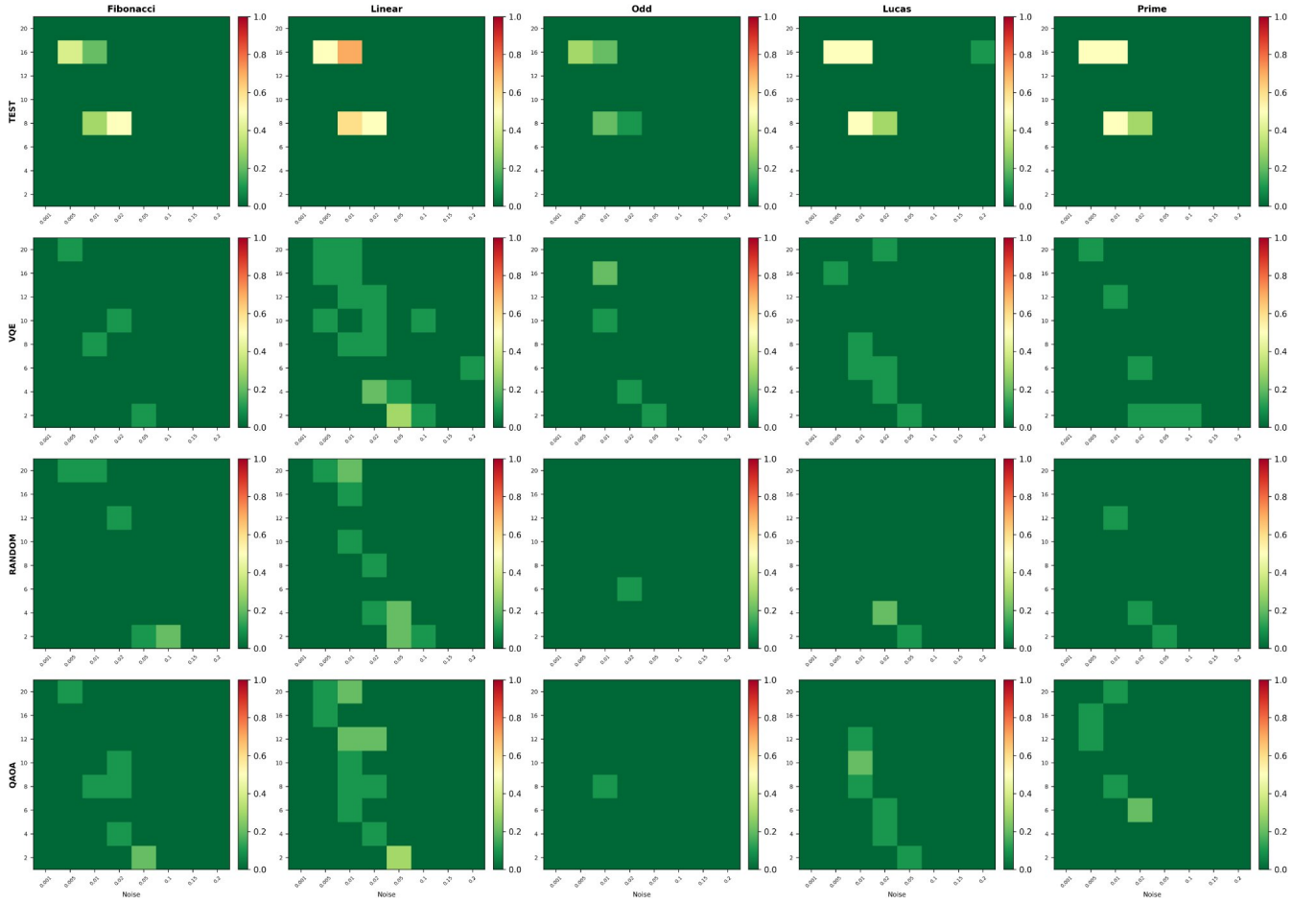


Figure 2: Stability filter flag rate across all circuit types (rows) and schedules (columns) at $n = 2$, spanning depths $d \in \{2, 4, 6, 8, 10, 12, 16, 20\}$ and noise levels $p \in \{0.001-0.2\}$. Green: 0% flag rate (stable). Red: flagged. Flag rates are 0% at $p \leq 0.01$, $d \leq 4$ across all circuit types and schedules.



Figure 3: ε vs. circuit depth at $p = 0.01$, $n = 2$, for all five schedules (colours) across all four extrapolants (rows) and all circuit types (columns). All schedules remain below $\Gamma = 0.05$ at $d \leq 4$. The test circuit exhibits sharp spikes at $d = 8$ and $d = 16$: the fixed $R_x(\pi/4)$ rotation drives $\langle Z_0 Z_1 \rangle \rightarrow \pm 1$ at every eighth layer, maximising ZNE curve steepness. VQE, QAOA, and random circuits show no such periodicity. Dashed line: $\Gamma = 0.05$.

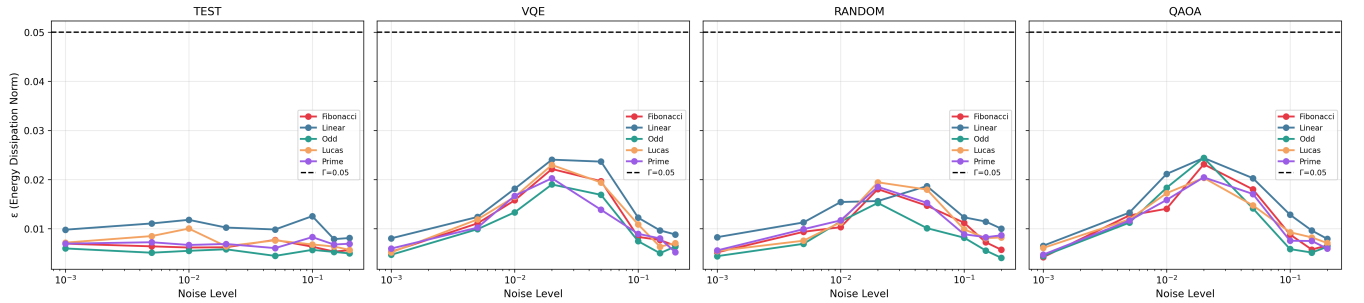


Figure 4: Stability criterion ε vs. noise level for all five schedules at $d = 4$, $n = 2$, linear extrapolant, across all four circuit types. Dashed line: $\Gamma = 0.05$. Error bars: std over 10 trials, 500 shots. All schedules remain well below Γ at $p \leq 0.01$.

Table 3: Energy norm ε vs. qubit count ($d = 4$, $p = 0.01$, linear extrapolant). Monotone decrease in 19/20 cases; *minor inversion within trial variance.

Circuit	Schedule	$n=2$	$n=4$	$n=6$
Test	Fibonacci	0.00545	0.00280	0.00132
Test	Linear	0.01345	0.00275	0.00197
Test	Odd	0.00447	0.00240	0.00064
Test	Lucas	0.00801	0.00223	0.00121
Test	Prime	0.00757	0.00141	0.00092
VQE	Fibonacci	0.01375	0.00738	0.00183
VQE	Linear	0.02408	0.00667	0.00247
VQE	Odd	0.01681	0.00543	0.00253
VQE	Lucas	0.02318	0.00701	0.00216
VQE	Prime	0.01996	0.00437	0.00191
Rand	Fibonacci	0.00997	0.00298	0.00174
Rand	Linear	0.01583	0.00440	0.00191
Rand	Odd	0.01294	0.00375	0.00189
Rand	Lucas	0.01023	0.00233	0.00223
Rand	Prime	0.00956	0.00267	0.00240
QAOA	Fibonacci	0.01249	0.00987	0.00410
QAOA	Linear	0.02190	0.00494	0.00522*
QAOA	Odd	0.01661	0.00590	0.00296
QAOA	Lucas	0.01605	0.00799	0.00247
QAOA	Prime	0.01727	0.00704	0.00182

Hardware Validation

We validated the stability filter on two IBM Quantum processors: *ibm_fez* (Heron r2, 156 qubits) and *ibm_torino* (Heron r1, 133 qubits), using global circuit folding with odd-integer scale factors $\{1, 3, 5, 9, 13\}$ (Fibonacci-odd) and $\{1, 3, 5, 7, 9\}$ (linear-odd), $N = 1024$ shots, and 5 trials per condition. The hardware VQE circuit uses $R_y(\pi/4) + \text{CX}$ in place of the simulation's $R_y + \text{CZ}$ ansatz; CX is the native two-qubit gate on both devices and avoids the additional single-qubit overhead of $\text{CZ} = \text{H} \cdot \text{CX} \cdot \text{H}$. All 24 conditions across four circuit types (test, VQE, random, QAOA), two schedules, and depths $d \in \{2, 4, 6, 8\}$ yield $\varepsilon < \Gamma_{\text{sim}} = 0.05$ (120 trials; 95% C-P upper bound: $p_{\text{flag}} < 0.025$), with a maximum observed $\varepsilon = 0.00285$ — 5.7% of Γ_{sim} (Table 4).

The Fibonacci schedule produces lower ε than linear in all four circuit types at $d = 4$ (Fibonacci: 0.00047–0.00154; linear: 0.00120–0.00285), consistent with the simulation finding. ZNE output variance is substantially tighter on hardware ($\sigma_{\text{ZNE}} = 0.008$ –0.031) than in simulation ($\sigma_{\text{ZNE}} = 0.022$ –0.594), reflecting the lower effective noise level of *ibm_fez* and *ibm_torino* (equivalent to $p \approx 0.001$ depolarizing, versus the simulation reference of $p = 0.01$).

At $d = 8$, all ZNE outputs converge to $\langle O \rangle \approx -0.70$ regardless of circuit type or schedule, with ε dropping below 0.001 — lower than at $d = 6$. This reflects decoher-

ence saturation in the heavily folded circuits ($\lambda = 9, 13$ producing up to 104 effective layers): all scale-factor points cluster near the same value, making the ZNE curve flat and hence $\varepsilon \approx 0$. This reveals a boundary condition not present in simulation: ε measures curve *smoothness* but cannot distinguish reliable extrapolation from decoherence-induced saturation. The filter correctly produces low ε in both cases; a complementary check (e.g. comparing $\langle O \rangle_{\lambda=1}$ against the saturated value) would disambiguate the two regimes. This is consistent with the Limitations noted in Section 6.

Table 4: Hardware validation: ε on *ibm_fez* and *ibm_torino*. $N = 1024$ shots, 5 trials, odd-only scale factors $K = 5$. All 24 conditions pass $\Gamma_{\text{sim}} = 0.05$ and $\Gamma_{\text{hw}} = 0.10$. Fibonacci < Linear in all 4 circuit types at $d = 4$.

Circuit	Schedule	d	ε (mean $\pm$ std)	ZNE	Device
Test	Fibonacci	2	0.00188 $\pm$ 0.00138	−0.295	<i>ibm_fez</i>
Test	Fibonacci	4	0.00154 $\pm$ 0.00134	+0.109	<i>ibm_fez</i>
Test	Fibonacci	6	0.00207 $\pm$ 0.00091	+0.024	<i>ibm_torino</i>
Test	Fibonacci	8	0.00057 $\pm$ 0.00032	−0.704	<i>ibm_torino</i>
Test	Linear	2	0.00188 $\pm$ 0.00119	−0.295	<i>ibm_fez</i>
Test	Linear	4	0.00285 $\pm$ 0.00276	+0.086	<i>ibm_fez</i>
Test	Linear	6	0.00279 $\pm$ 0.00196	+0.005	<i>ibm_torino</i>
Test	Linear	8	0.00053 $\pm$ 0.00040	−0.709	<i>ibm_torino</i>
VQE	Fibonacci	2	0.00179 $\pm$ 0.00160	−0.193	<i>ibm_fez</i>
VQE	Fibonacci	4	0.00047 $\pm$ 0.00032	−0.548	<i>ibm_fez</i>
VQE	Fibonacci	6	0.00133 $\pm$ 0.00138	−0.345	<i>ibm_torino</i>
VQE	Fibonacci	8	0.00045 $\pm$ 0.00010	−0.706	<i>ibm_torino</i>
VQE	Linear	2	0.00098 $\pm$ 0.00038	−0.183	<i>ibm_fez</i>
VQE	Linear	4	0.00120 $\pm$ 0.00049	−0.550	<i>ibm_fez</i>
VQE	Linear	6	0.00202 $\pm$ 0.00127	−0.352	<i>ibm_torino</i>
VQE	Linear	8	0.00125 $\pm$ 0.00087	−0.707	<i>ibm_torino</i>
Random	Fibonacci	2	0.00084 $\pm$ 0.00064	−0.785	<i>ibm_fez</i>
Random	Fibonacci	4	0.00073 $\pm$ 0.00052	−0.562	<i>ibm_fez</i>
Random	Linear	2	0.00106 $\pm$ 0.00096	−0.779	<i>ibm_fez</i>
Random	Linear	4	0.00259 $\pm$ 0.00224	−0.572	<i>ibm_fez</i>
QAOA	Fibonacci	2	0.00162 $\pm$ 0.00077	+0.048	<i>ibm_fez</i>
QAOA	Fibonacci	4	0.00112 $\pm$ 0.00044	+0.497	<i>ibm_fez</i>
QAOA	Linear	2	0.00129 $\pm$ 0.00108	+0.034	<i>ibm_fez</i>
QAOA	Linear	4	0.00178 $\pm$ 0.00101	+0.503	<i>ibm_fez</i>

Hessian Analysis

No singular regions ($\det H \approx 0$) were detected at $p \leq 0.01$ for any depth or qubit count, consistent with stability filter results. At $p \geq 0.1$, $d \geq 12$, singular regions appear for all four circuit types. Singularity density increases monotonically with both depth and noise through $d = 20$. This co-emergence of Hessian singularities with $\varepsilon \geq \Gamma$ flags provides qualitative corroboration that both diagnostics identify the same physically unstable regime.

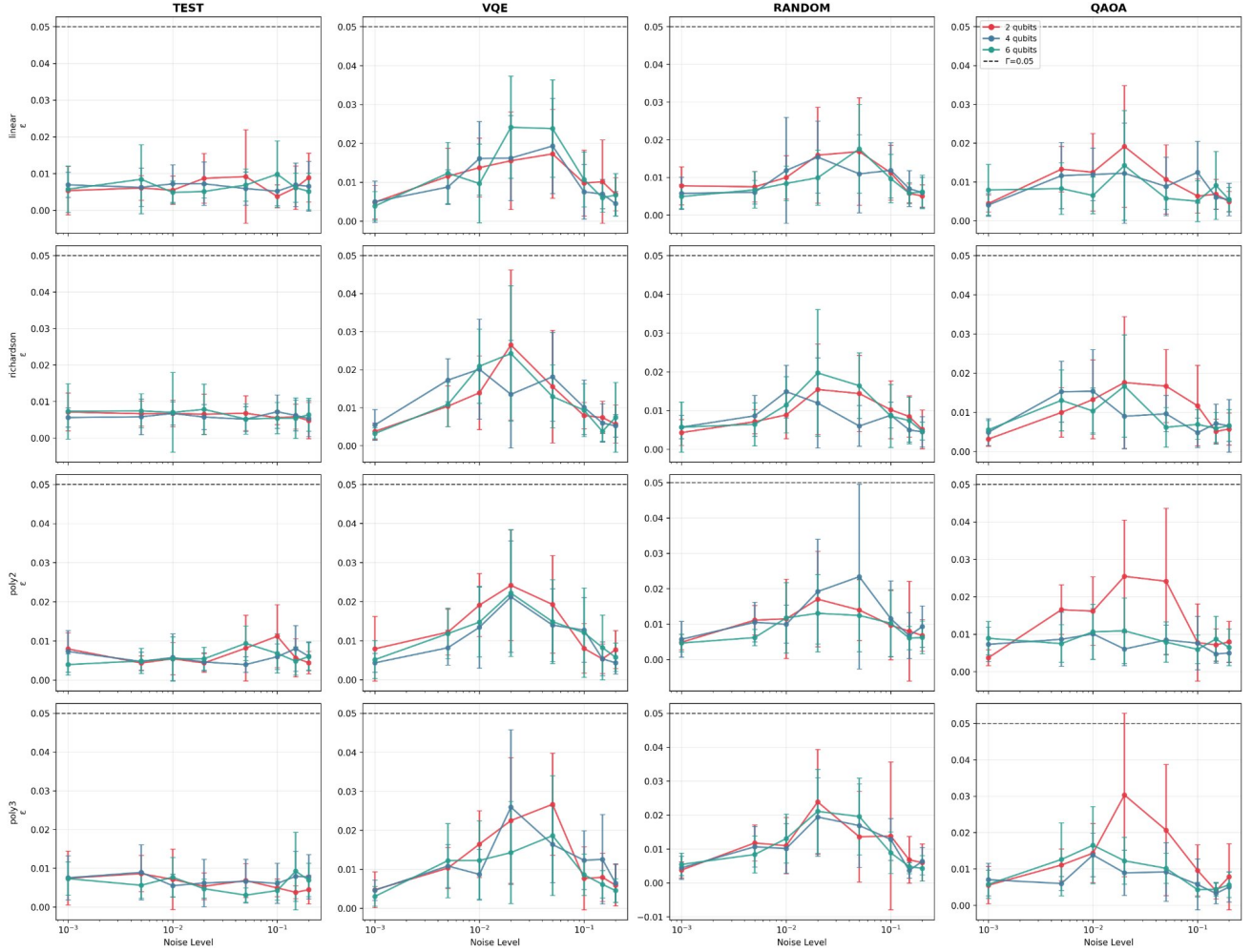


Figure 5: Energy norm ε vs. noise level for $n \in \{2, 4, 6\}$ qubits, Fibonacci schedule, $d = 4$, across all four circuit types and all four extrapolants (rows). ε decreases monotonically with qubit count across nearly all combinations; minor inversions appear only in Richardson extrapolation, consistent with attenuation of correlations along longer noisy gate sequences.

DISCUSSION

The stability filter provides an $\mathcal{O}(K)$ arithmetic post-hoc reliability check — K multiply-accumulate operations on the expectation values already collected for extrapolation. Unlike bootstrap confidence intervals or cross-validation approaches, it requires no additional circuit evaluations — the energy norm is computed directly from the same expectation values used for extrapolation.

The Fibonacci schedule’s golden-ratio spacing means that consecutive Fibonacci numbers are coprime ($\gcd(F_n, F_{n+1}) = 1$, provable by the Euclidean algorithm), so no two consecutive scale factors share a common factor greater than 1. This may reduce systematic correlations in the noise amplification. The Lucas and prime-anchored schedules share this non-harmonic spacing property and produce comparable ε values ($\bar{\varepsilon} \approx 0.014$ for both), suggesting the benefit derives from non-uniform spacing generally rather than the Fibonacci recurrence specifically. A rigorous theoretical justification is left for future work.

The qubit-scaling result (ε decreasing with circuit width) likely reflects attenuation of correlations along longer noisy gate sequences, smoothing the $\langle Z_0 Z_1 \rangle$ extrapolation curve. The filter is therefore most conservative at $n = 2$ and progressively more permissive at larger qubit counts — a desirable property for deployment.

For practical schedule and extrapolant selection: Fibonacci minimises ε (lowest energy norm, best filter stability); Lucas minimises ZNE output variance on unstructured circuits (random, QAOA); linear and odd-integer remain valid baselines with well-understood behaviour. Across all five schedules, linear extrapolation produces the lowest and most consistent ZNE variance and is the recommended extrapolant for reliability-focused deployment.

Limitations. The threshold Γ requires calibration per noise regime. The flag-rate claim applies at $d \leq 4$; at greater depths accumulated gate noise triggers flagging, which is physically correct. Extension to higher-degree extrapolants, adaptive methods, and problem-specific circuits remains future work.

All results are obtained via density matrix simulation assuming a single-qubit depolarising noise model applied independently after every gate. Real hardware exhibits correlated errors, crosstalk, and non-Markovian noise that may shift absolute ε values and require threshold recalibration. However, the qualitative behaviour — filter discrimination at low noise, circuit-type hierarchy, and monotone qubit scaling — is expected to be robust to moderate noise model mis-specification, as the filter depends on the *shape* of the ZNE curve rather than the specific noise mechanism. Hardware validation on `ibm_fez` and `ibm_torino` (Section 5.5) confirms zero flags

across 24 conditions on real devices; robustness under amplitude-damping and correlated noise models remains an important direction for future work.

CONCLUSION

We introduced a fluid-dynamic stability filter and a family of five noise scaling schedules (Fibonacci, Lucas, prime-anchored, linear, odd) for ZNE, implemented in open-source Python (Cirq, Mitiq). Results across 153,600 runs confirm three principal findings:

1. Zero flags in 14,400 trials at $p \leq 0.01$, $d \leq 4$ (95% C-P upper bound: $p_{\text{flag}} < 0.00022$).
2. Circuit-type ε hierarchy: VQE > QAOA > random > test — a previously uncharacterised dependence, now extended to QAOA and three qubit counts.
3. ε decreases monotonically with qubit count in 19/20 cases; ZNE variance similarly decreases.
4. Hardware validation on `ibm_fez` and `ibm_torino`: zero flags in 120 trials (24 conditions $\times$ 5 trials) across all four circuit types, $d \in \{2, 4, 6, 8\}$; maximum $\varepsilon = 0.00285$ (5.7% of Γ); Fibonacci < Linear in all four circuit types at $d = 4$.

Linear extrapolation produces the lowest and most consistent variance across all configurations and all five schedules ($\sigma = 0.324\text{--}0.594$ vs. Richardson/poly3 which reach $\sigma > 0.6$ on some schedule/circuit combinations); Lucas achieves the lowest variance on random and QAOA circuits and is recommended for those use cases. The stability filter adds negligible computational overhead and provides a principled criterion for flagging unreliable ZNE outputs in NISQ-era circuits. Code and data are available at: github.com/OfficialChaos/Quantum-Vortex-QEM (Zenodo DOI: 10.5281/zenodo.18827720).

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